

# Product vs Entangled states

$$\rho = \sum_{i,j=0}^{d-1} \underbrace{\langle i| \rho |j \rangle}_{\text{matrix}} \underbrace{|i\rangle\langle j|}_{\text{op.}}$$

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$$\langle i|X|j\rangle = |i\rangle\langle j| \otimes |f(X)|$$

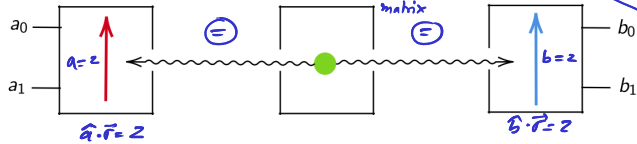
$$X|a\rangle = -|a\rangle$$

Independent

$$\mathcal{B}_2 = \{|0\rangle, |1\rangle\}$$

$$\langle (a,b) = \langle 4 | (\hat{a} \cdot \vec{\sigma}) (\hat{b} \cdot \vec{\sigma}) | 4 \rangle$$

$$\in [-1, 1]$$



Product (factorizable) states-vector:  $\frac{1}{\sqrt{2}} (|0\rangle_a |0\rangle_b - |1\rangle_a |0\rangle_b) = |4\rangle_{ab} = \frac{1}{\sqrt{2}} (|0\rangle_a - |1\rangle_a) |0\rangle_b$   
 $Z \otimes Z |4\rangle = \frac{1}{\sqrt{2}} (|0\rangle |0\rangle + |1\rangle |0\rangle)$

$a = b = z$

|               |   |   |   |   |   |     |
|---------------|---|---|---|---|---|-----|
| a-measurement | 0 | 1 | 1 | 0 | 1 | ... |
| b-measurement | 0 | 0 | 0 | 0 | 0 | ... |

NO correlation

$$Z|0\rangle_b = |0\rangle_b$$

$$Z|1\rangle = -|1\rangle$$

$a=x$   
 $b=z$

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

same diff  
 $\langle 4 | X \otimes Z | 4 \rangle = \langle -1 | -1 \rangle_a \langle 0 | z | 0 \rangle_b = (-1)(1) = -1$

$$\langle 4 | Z \otimes Z | 4 \rangle = 0$$

Entangled (non-factorizable) states-vector:  $|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle_a |1\rangle_b - |1\rangle_a |0\rangle_b)$

$a = b = z$

|               |   |   |   |   |   |     |
|---------------|---|---|---|---|---|-----|
| a-measurement | 0 | 1 | 1 | 0 | 1 | ... |
| b-measurement | 1 | 0 | 0 | 1 | 0 | ... |

totally Random

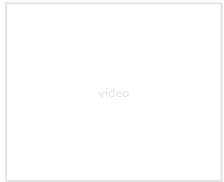
anti

PERFECT correlation

$$Z \otimes Z |4\rangle = \frac{1}{\sqrt{2}} (-|0\rangle |1\rangle + |1\rangle |0\rangle) = -|4\rangle$$

$$\langle 4 | Z \otimes I | 4 \rangle = 0$$

$$\langle 4 | Z \otimes Z | 4 \rangle = -\langle 4 | 4 \rangle = -1$$



# Representing a single-qubit system (RECALL)

True

ket

State vectors in  $\mathcal{H}_2$ :  $|\psi\rangle = \left[ \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} e^{i\phi} |1\rangle \right] e^{i\alpha}$   
 where  $\theta \in [0, \pi]$  and  $\phi \in [0, 2\pi)$ .

Operators on  $\mathcal{H}_2$ :  $A = \sum_{k=0}^3 \gamma_k \sigma_k$  where  $\gamma_k = \text{tr}(\sigma_k A) \in \mathbb{C}$ .

$$[\sigma_k, A] = \text{tr}(\sigma_k^\dagger A)$$

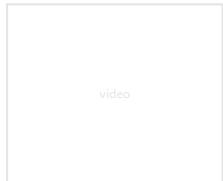
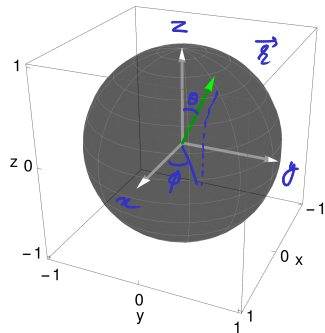
| Vector-basis                               | Operator-basis                                                                                           |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------|
| $\mathcal{B}_2 = \{ 0\rangle,  1\rangle\}$ | $\mathfrak{B} = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3\} = \{I, \underbrace{X, Y, Z}_{\vec{\sigma}}\}$ |

Single qubit state:  $\rho = \frac{I + \vec{r} \cdot \vec{\sigma}}{2}$  where  $|\vec{r}| \leq 1$ .  $0 \leq \rho$   
 $= 1/4 \times 4$

Bloch vector:  $\vec{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle) \in \mathbb{R}^3$

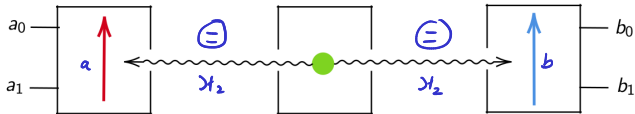
$$\langle x \rangle = \text{tr}(x \rho) \quad \text{Born Rule}$$

Bloch Ball



# Representing a two-qubit system

$$\rho = \sum_{ij} \underbrace{\langle i f | \rho | j g \rangle}_{p_{if,jg}} |i f \rangle \langle j g|$$



$$\begin{bmatrix} \rho_{00,00} & \dots & \rho_{00,11} \\ \vdots & & \vdots \\ \rho_{11,00} & \dots & \rho_{11,11} \end{bmatrix}_{4 \times 4}$$

density matrix

Basis Encoding  
00  $\rightarrow$  |00>

$\dim = 2 \leftarrow$  no of qubits  
 $\dim \mathcal{H} = 4$

Vector-basis of  $\mathcal{H}_2 \otimes \mathcal{H}_2$

$$\mathcal{B} = \{|0\rangle_a, |1\rangle_a\} \otimes \{|0\rangle_b, |1\rangle_b\}$$

$$= \{ |00\rangle_{ab}, |01\rangle_{ab}, |10\rangle_{ab}, |11\rangle_{ab} \}, \text{ where } |00\rangle_{ab} = |0\rangle_a \otimes |0\rangle_b$$

$$|01\rangle_{ab},$$

$$|10\rangle_{ab},$$

$$|11\rangle_{ab} \}, \text{ where}$$

$$|ij\rangle_{ab} = |i\rangle_a \otimes |j\rangle_b \text{ and}$$

$$|\psi\rangle_{ab} = \sum_{i,j} \alpha_{ij} |ij\rangle_{ab}, \quad \alpha_{ij} = \langle ij | \psi \rangle$$

prob amp

$$1024 = 2^{10}$$

Operator-basis

$$\mathcal{B} = \{ \sigma_0, \sigma_1, \sigma_2, \sigma_3 \}_a \otimes \{ \sigma_0, \sigma_1, \sigma_2, \sigma_3 \}_b$$

$$= \{ II, = I \otimes I$$

$$XI, YI, ZI,$$

$$IX, IY, IZ,$$

$$XX, XY, XZ,$$

$$YX, YY, YZ,$$

$$ZX, ZY, ZZ \}$$

a-qubit spin vector

b-qubit spin vector

ab-correlators

$$A = \sum_{k,l} \gamma_{kl} \sigma_k \otimes \sigma_l,$$

$$\gamma_{kl} = \text{tr}(\sigma_k \otimes \sigma_l A)$$

TB

$$= \{ |i f \rangle \langle j g| \}$$

$$= \{ |i f \rangle \langle j g| \}$$

$$X_a = X \otimes I$$

$$\vec{r}_a = (\langle x_a \rangle, \langle y_a \rangle, \langle z_a \rangle)$$

$$\vec{r}_b$$

$$(\vec{a} \cdot \vec{\sigma})(\vec{b} \cdot \vec{\sigma})$$

video